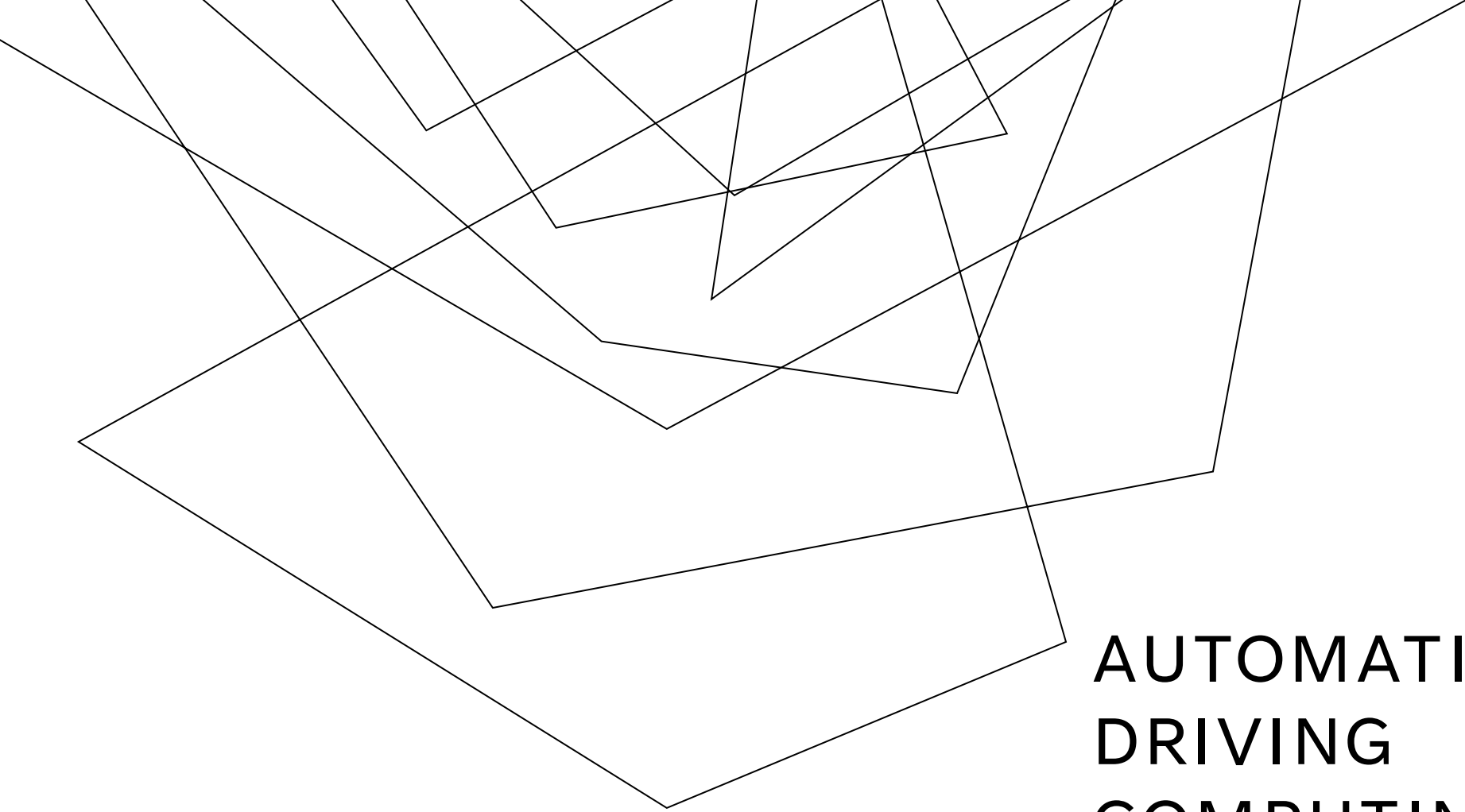


A series of thin, black, overlapping lines forming various geometric shapes like triangles and polygons, creating a complex, abstract pattern in the upper left portion of the slide.

AUTOMATIC DRIVING COMPUTING

6. PERCEPTION (4)





지난 시간 복습

OBJECT DETECTION (2)

1. R-CNN inference

2. Fast R-CNN





목차

목차

1. Faster R-CNN

2. Feature pyramid network



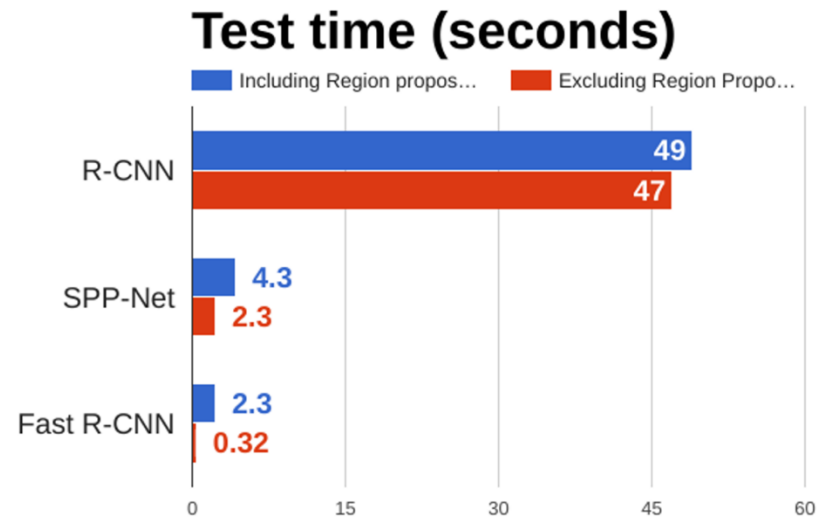
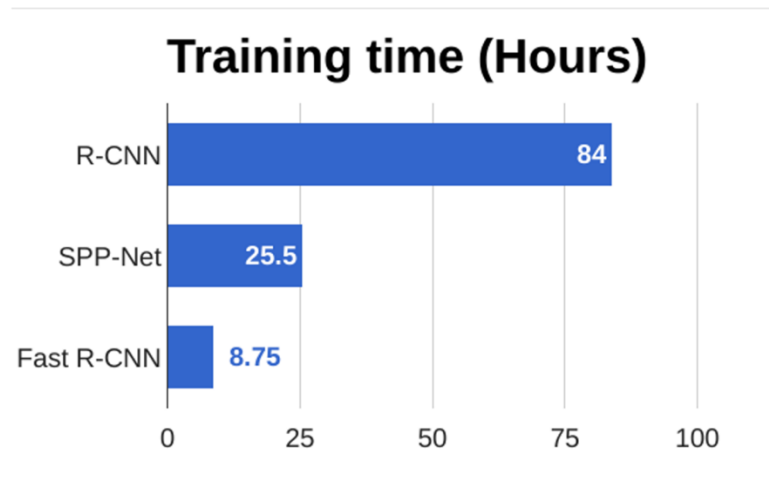
An abstract graphic design featuring two thin, dark gray lines that intersect on a light gray background. One line is oriented diagonally from the top-left towards the bottom-right, while the other is oriented from the top-right towards the bottom-left. The intersection point is located in the upper-left quadrant of the image.

FASTER R-CNN

LIMITATION OF FAST R-CNN

Image의 region proposal 얻기 위해 돌리는 알고리즘 (e.g., Selective Search)
의 실행 시간이 bottleneck으로 작용

해당 알고리즘은 CPU에서 돌아가는 고전 알고리즘임



FASTER R-CNN

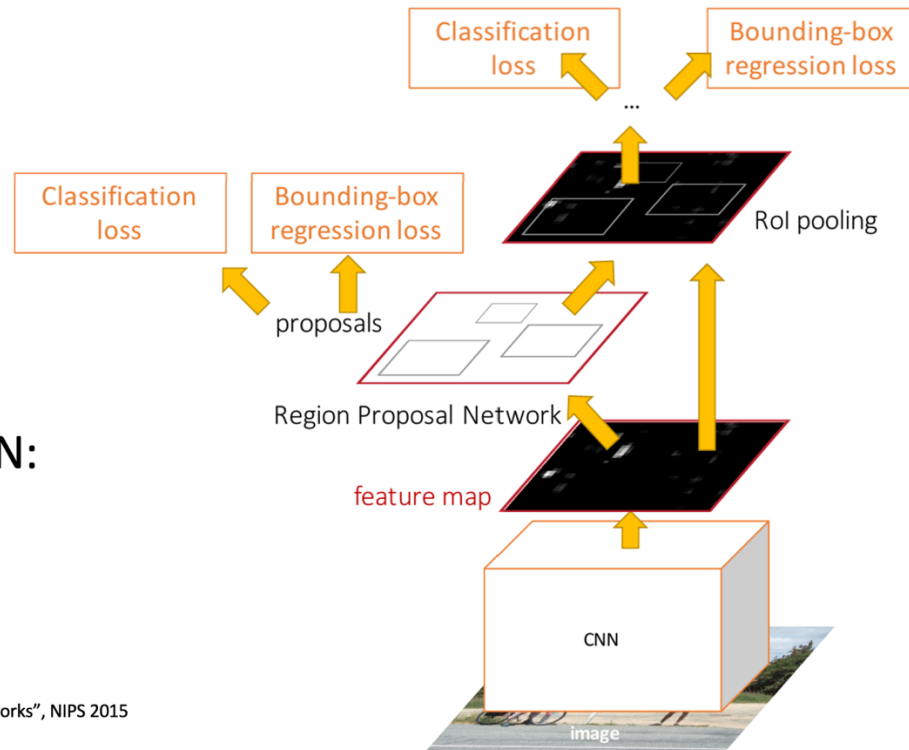
NeurIPS 2015

Solution : Region proposal을 고전 알고리즘에 의존 X, Learnable하게 만들자

Region proposal network (RPN) - CNN feature로부터 proposal 예측

Insert **Region Proposal Network (RPN)** to predict proposals from features

Otherwise same as Fast R-CNN:
Crop features for each proposal, classify each one



Ren et al, "Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks", NIPS 2015
Figure copyright 2015, Ross Girshick; reproduced with permission



REGION PROPOSAL NETWORK

Region Proposal Network (RPN)

Run backbone CNN to get
features aligned to input image



Input Image
(e.g. 3 x 640 x 480)

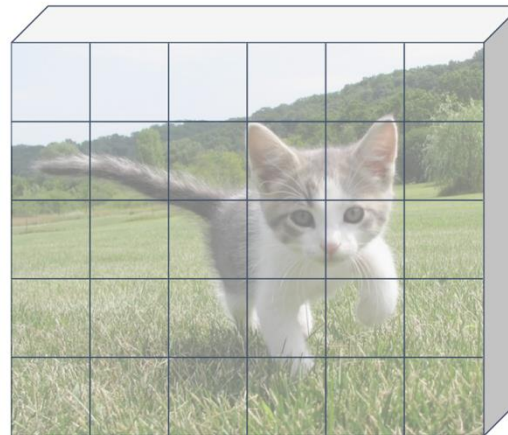


Image features
(e.g. 512 x 5 x 6)

Ren et al, "Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks", NIPS 2015



REGION PROPOSAL NETWORK

Region Proposal Network (RPN)

Run backbone CNN to get
features aligned to input image



Input Image
(e.g. 3 x 640 x 480)



Each feature corresponds
to a point in the input

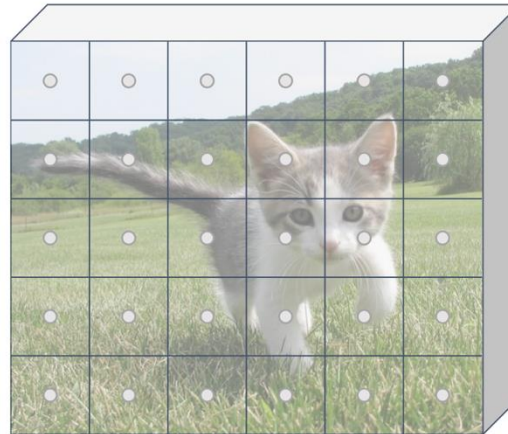


Image features
(e.g. 512 x 5 x 6)

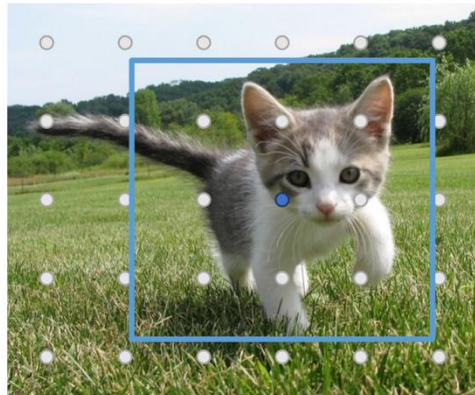
Ren et al, "Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks", NIPS 2015



REGION PROPOSAL NETWORK

Region Proposal Network (RPN)

Run backbone CNN to get
features aligned to input image



Input Image
(e.g. 3 x 640 x 480)



Each feature corresponds
to a point in the input

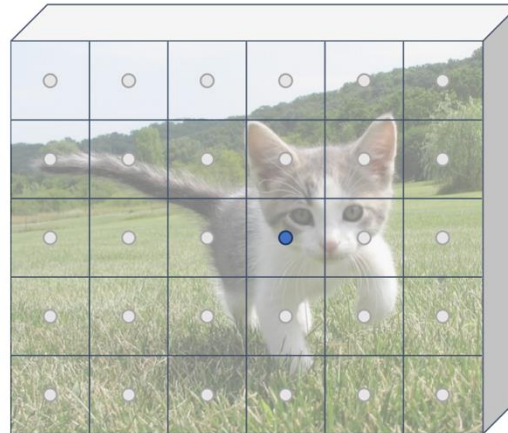


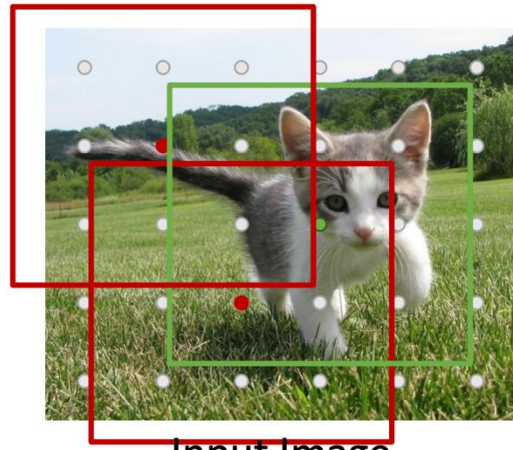
Image features
(e.g. 512 x 5 x 6)

Imagine an **anchor box**
of fixed size at each
point in the feature map

REGION PROPOSAL NETWORK

Region Proposal Network (RPN)

Run backbone CNN to get features aligned to input image



Input Image
(e.g. 3 x 640 x 480)

Each feature corresponds to a point in the input

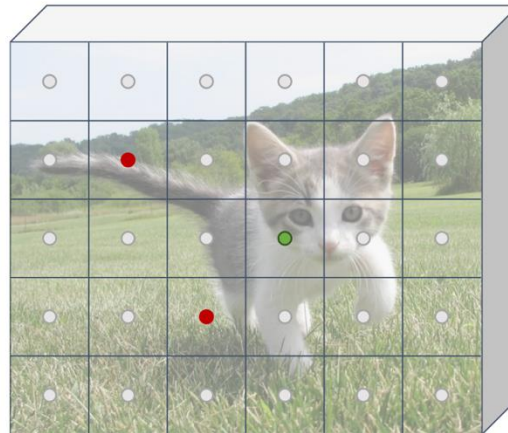


Image features
(e.g. 512 x 5 x 6)

Imagine an **anchor box** of fixed size at each point in the feature map

Classify each anchor as **positive (object)** or **negative (no object)**

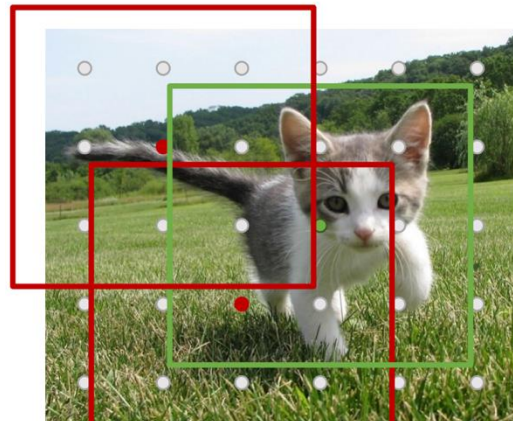
Ren et al, "Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks", NIPS 2015



REGION PROPOSAL NETWORK

Region Proposal Network (RPN)

Run backbone CNN to get features aligned to input image



Input Image
(e.g. 3 x 640 x 480)

CNN

Each feature corresponds to a point in the input

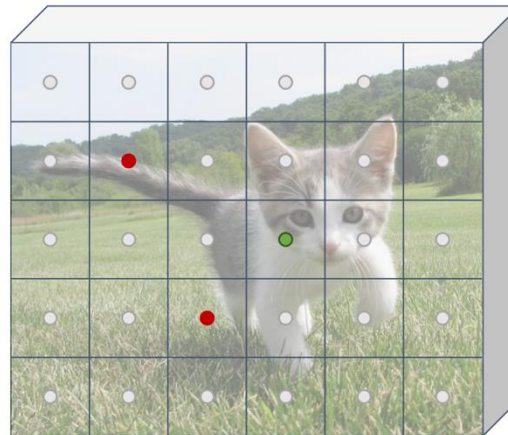


Image features
(e.g. 512 x 5 x 6)

Predict object vs not object scores for all anchors with a conv layer (512 input filters, 2 output filters)

Conv

Anchor is
object?
2 x 5 x 6

Classify each anchor as
positive (object) or
negative (no object)

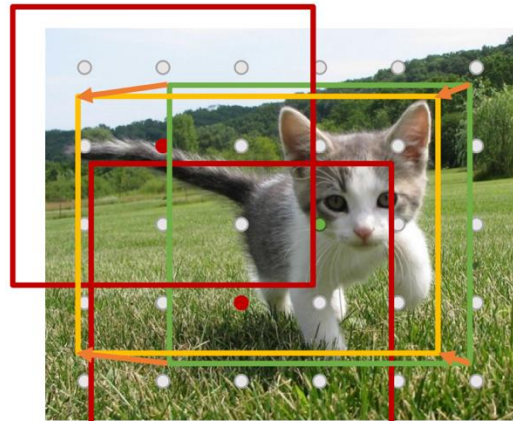
Ren et al, "Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks", NIPS 2015



REGION PROPOSAL NETWORK

Region Proposal Network (RPN)

Run backbone CNN to get features aligned to input image



Input Image
(e.g. 3 x 640 x 480)



Each feature corresponds to a point in the input

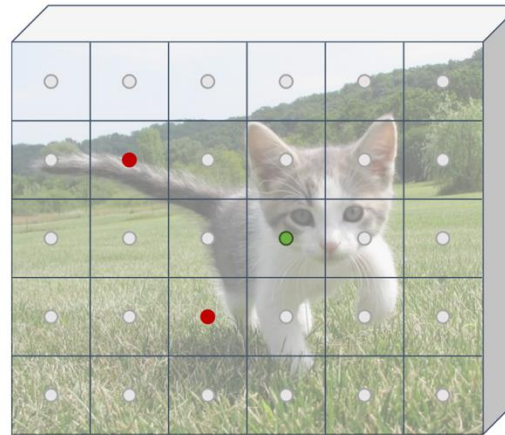


Image features
(e.g. 512 x 5 x 6)

For **positive anchors**, also predict a **transform** that converting the anchor to the **GT box** (like R-CNN)
Predict transforms with conv



→ Anchor is object?
2 x 5 x 6
→ Anchor transforms
4 x 5 x 6

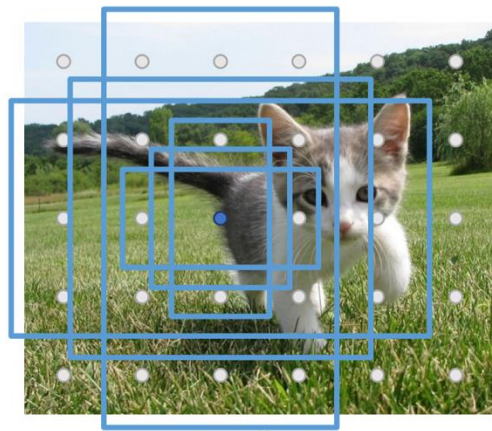
Classify each anchor as **positive (object)** or **negative (no object)**



REGION PROPOSAL NETWORK

Region Proposal Network (RPN)

Run backbone CNN to get features aligned to input image



Input Image
(e.g. 3 x 640 x 480)



Each feature corresponds to a point in the input

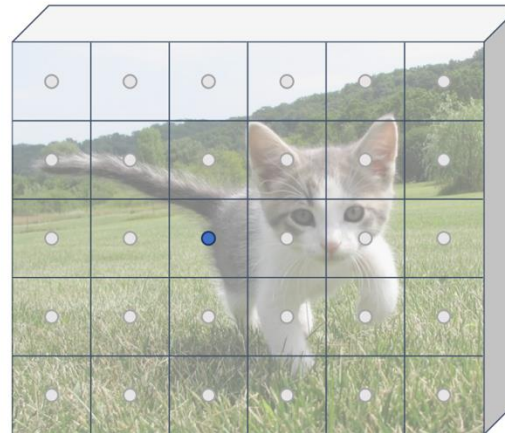


Image features
(e.g. 512 x 5 x 6)



In practice: Rather than using one anchor per point, instead consider K different anchors with different size and scale (here K = 6)

Anchor is object?
2K x 5 x 6

Anchor transforms
4K x 5 x 6

During training, supervised positive / negative anchors and box transforms like R-CNN

Positive anchors: ≥ 0.7 IoU with some GT box (plus highest IoU to each GT)

Negative anchors: < 0.3 IoU with all GT boxes. Don't supervised transforms for negative boxes.

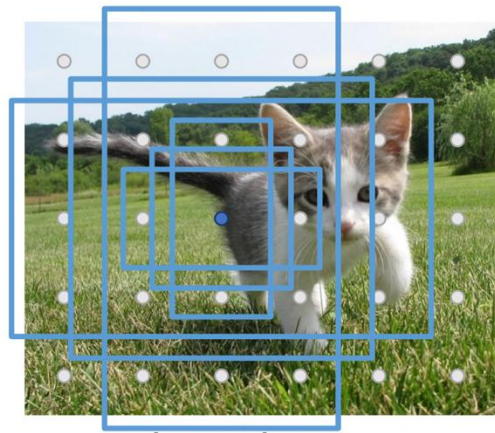
Neutral anchors: between 0.3 and 0.7 IoU with all GT boxes; ignored during training



REGION PROPOSAL NETWORK

Region Proposal Network (RPN)

Run backbone CNN to get features aligned to input image



Input Image
(e.g. 3 x 640 x 480)



Each feature corresponds to a point in the input

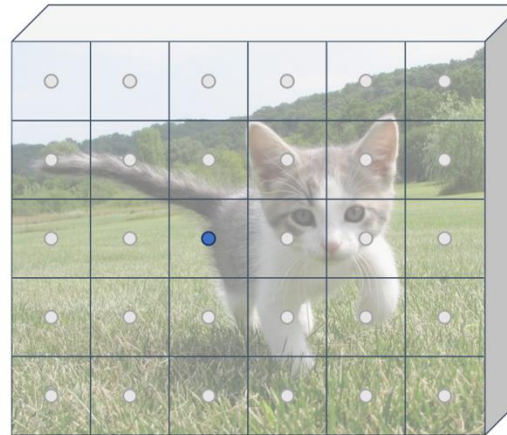


Image features
(e.g. 512 x 5 x 6)



In practice: Rather than using one anchor per point, instead consider K different anchors with different size and scale (here K = 6)

Anchor is object?
 $2K \times 5 \times 6$

Anchor transforms
 $4K \times 5 \times 6$

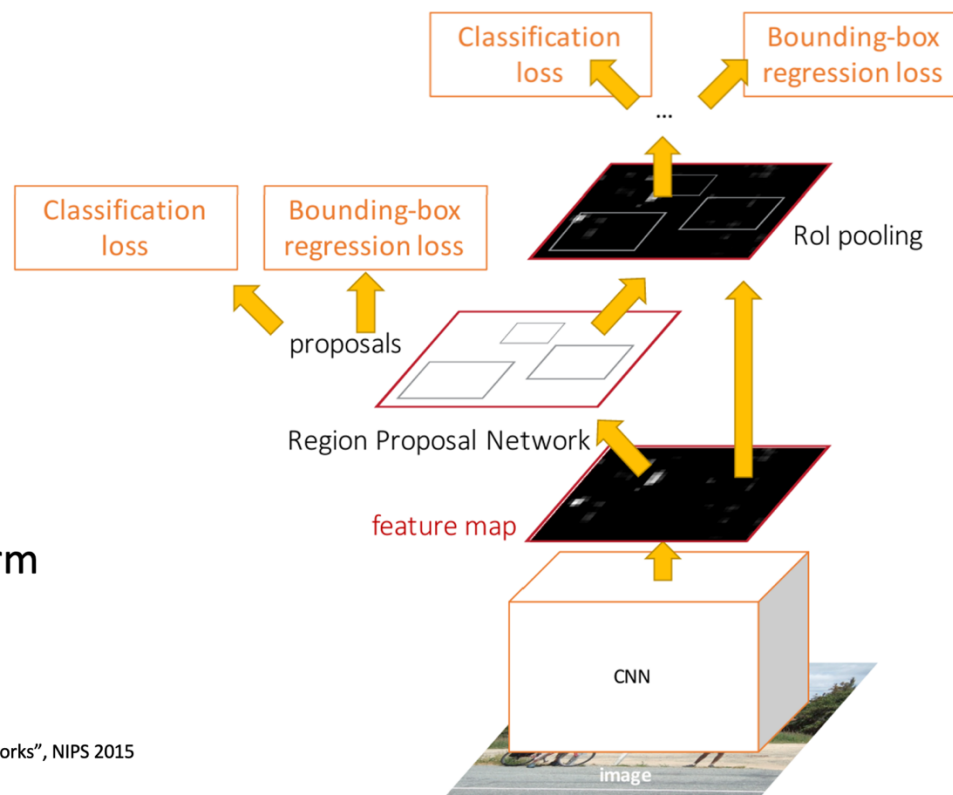
At test-time, sort all $K \times 5 \times 6$ boxes by their positive score, take top 300 as our region proposals



FASTER R-CNN

Jointly train with 4 losses:

1. **RPN classification:** anchor box is object / not an object
2. **RPN regression:** predict transform from anchor box to proposal box
3. **Object classification:** classify proposals as background / object class
4. **Object regression:** predict transform from proposal box to object box

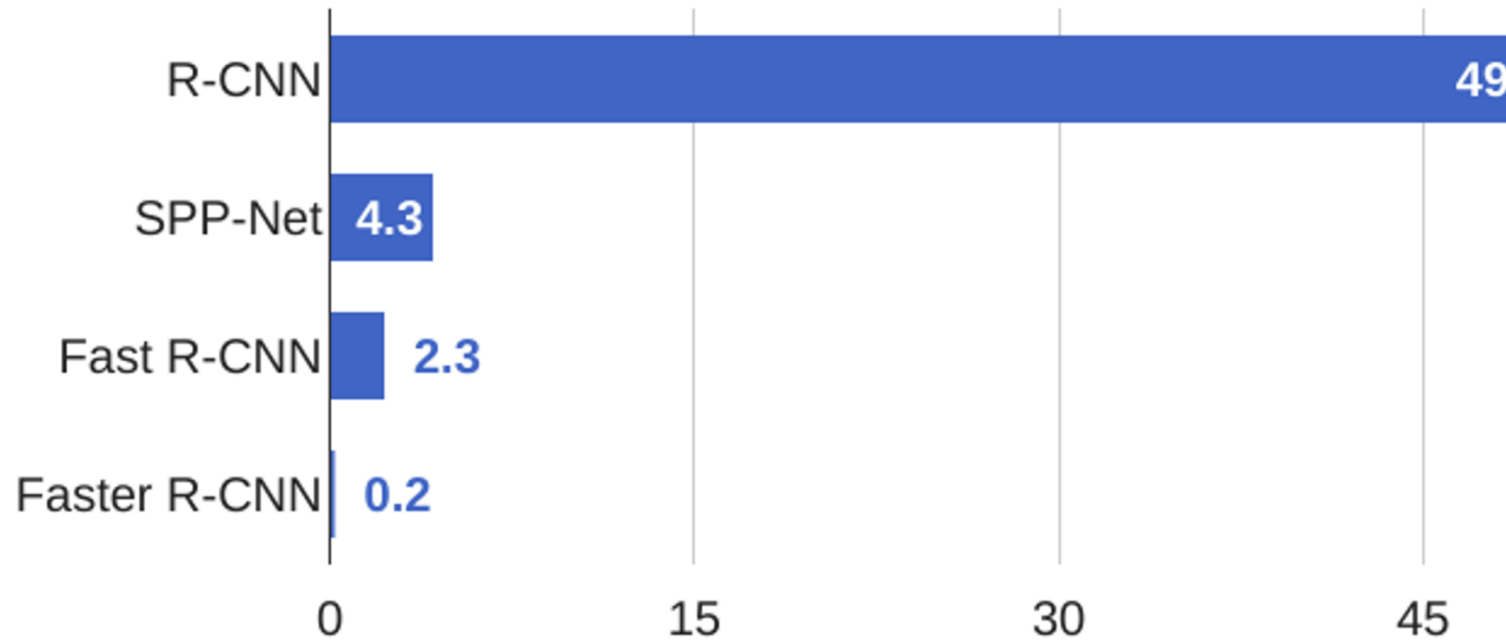


Ren et al, "Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks", NIPS 2015
Figure copyright 2015, Ross Girshick; reproduced with permission



FASTER R-CNN

R-CNN Test-Time Speed



FASTER R-CNN

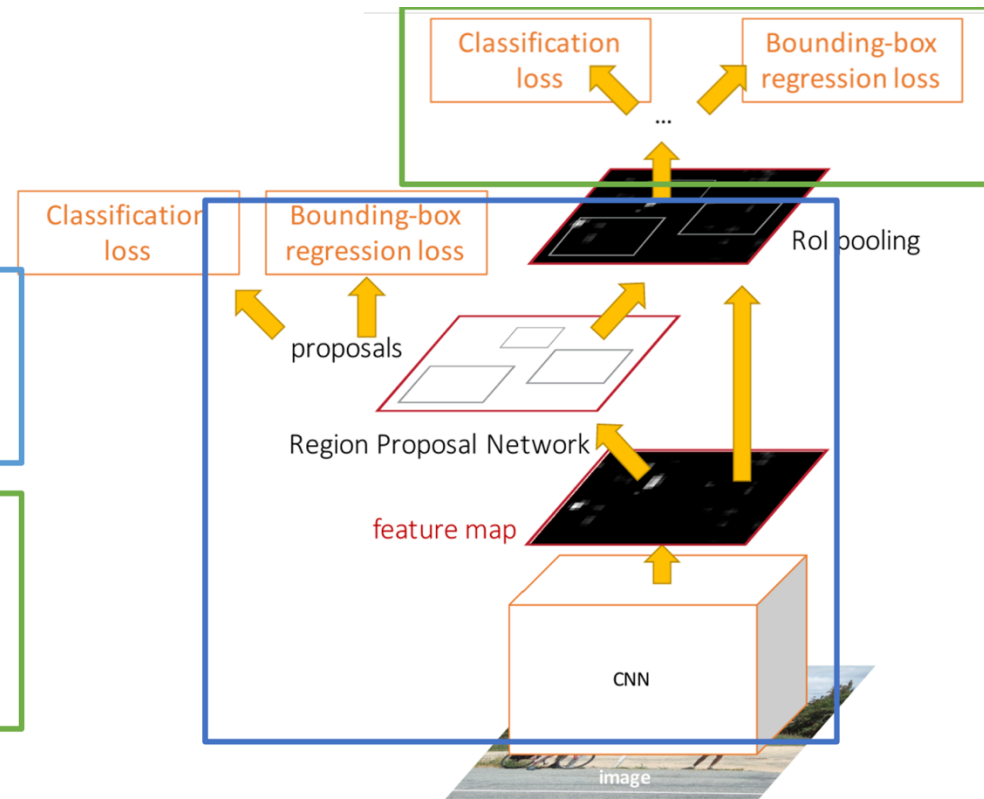
Faster R-CNN is a
Two-stage object detector

First stage: Run once per image

- Backbone network
- Region proposal network

Second stage: Run once per region

- Crop features: RoI pool / align
- Predict object class
- Prediction bbox offset



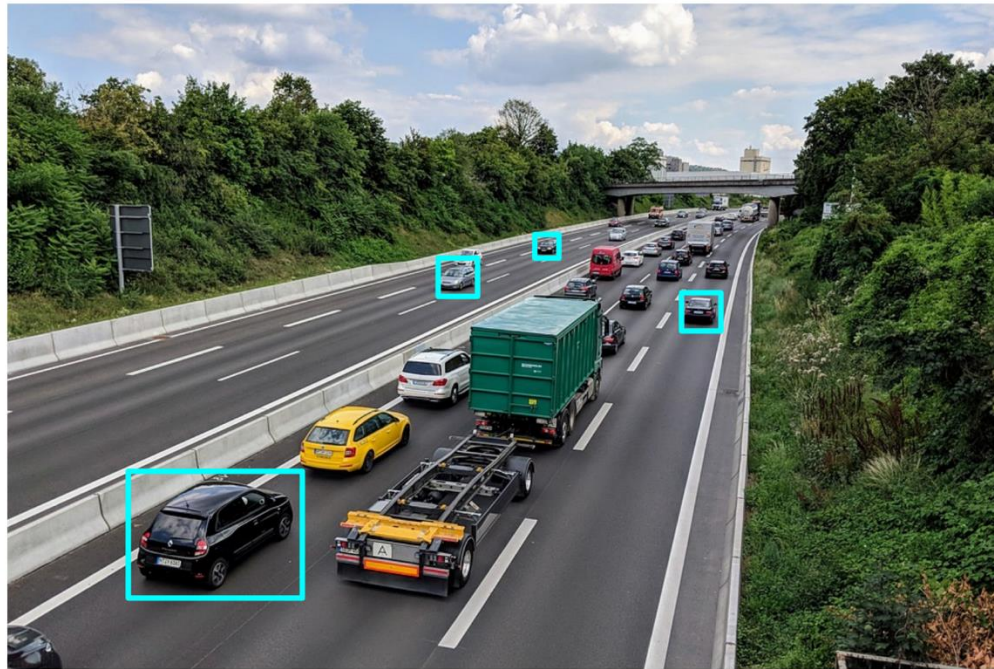
An abstract graphic design featuring two thin, dark gray lines that intersect on a light gray background. One line runs diagonally from the top-left towards the bottom-right, while the other runs from the top-right towards the bottom-left. The intersection point is located to the left of the text.

FEATURE PYRAMID NETWORK

SCALE ISSUE

Dealing with Scale

We need to detect objects of many different scales.
How to improve *scale invariance* of the detector?



[This image](#) is free for commercial use under the [Pixabay license](#)



SCALE ISSUE

작은 물체 (Small)에 대해서는 검출 성능이 많이 떨어짐.. Why?

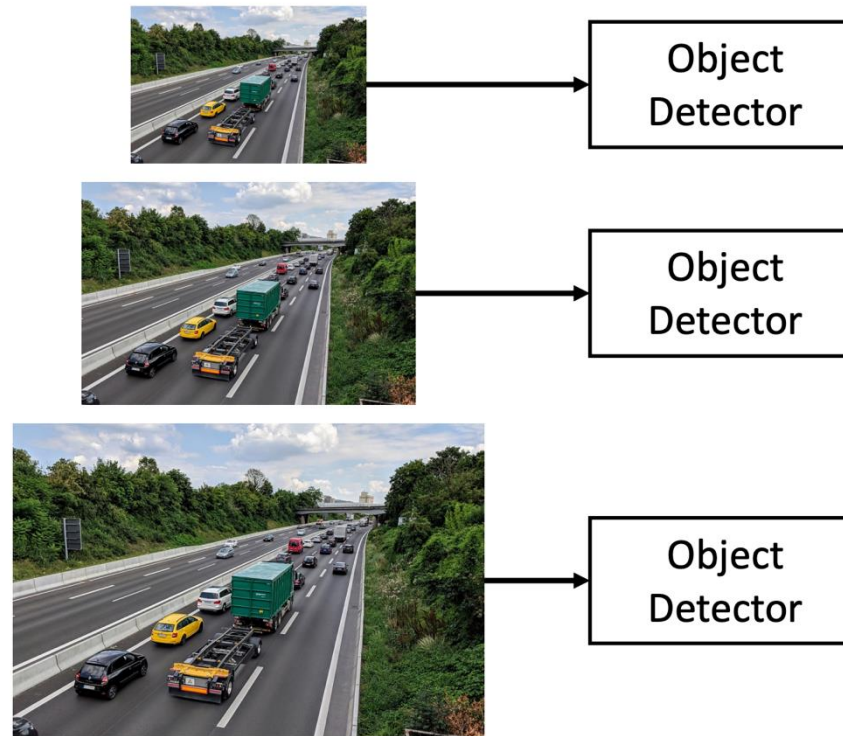
Method	data	Avg. Precision, IoU:			Avg. Precision, Area:		
		0.5:0.95	0.5	0.75	S	M	L
Fast [6]	train	19.7	35.9	-	-	-	-
Fast [24]	train	20.5	39.9	19.4	4.1	20.0	35.8
Faster [2]	trainval	21.9	42.7	-	-	-	-
ION [24]	train	23.6	43.2	23.6	6.4	24.1	38.3
Faster [25]	trainval	24.2	45.3	23.5	7.7	26.4	37.1



FEATURE PYRAMID NETWORK

Dealing with Scale: Image Pyramid

Classic idea: build an *image pyramid* by resizing the image to different scales, then process each image scale independently.



Lin et al, "Feature Pyramid Networks for Object Detection", ICCV 2017

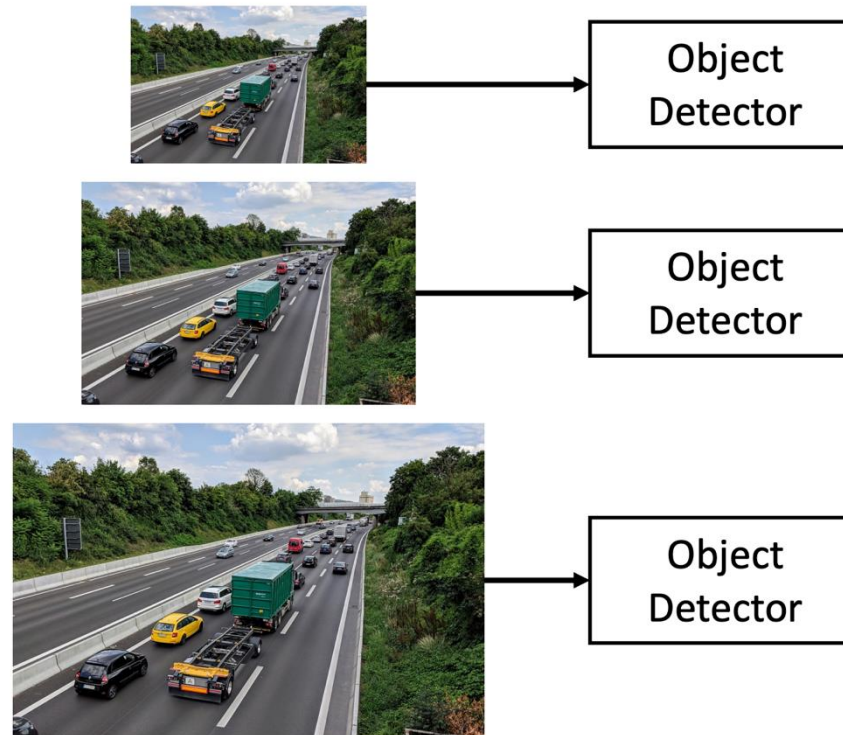


FEATURE PYRAMID NETWORK

Dealing with Scale: Image Pyramid

Classic idea: build an *image pyramid* by resizing the image to different scales, then process each image scale independently.

Problem: Expensive! Don't share any computation between scales



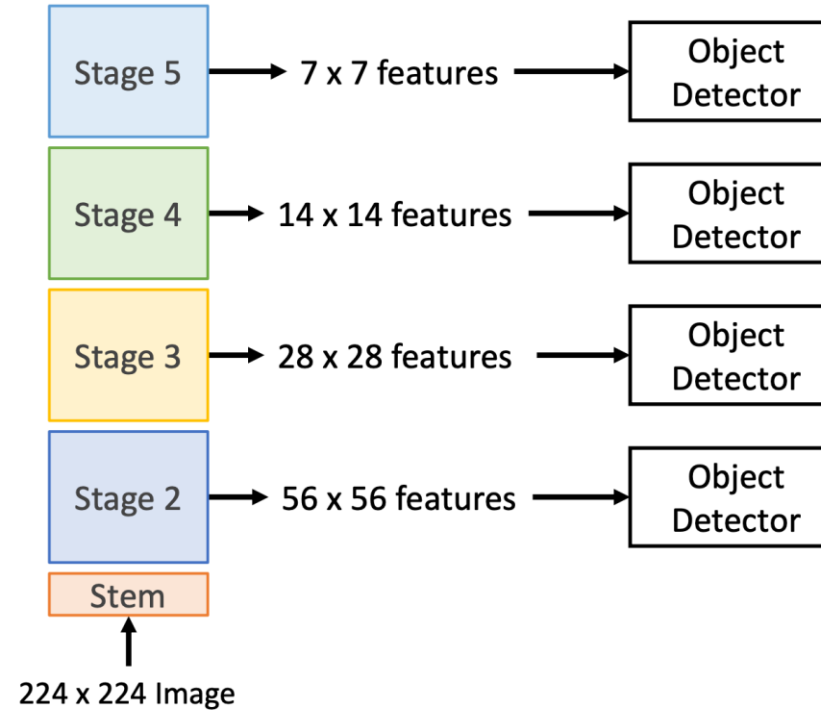
Lin et al, "Feature Pyramid Networks for Object Detection", ICCV 2017



FEATURE PYRAMID NETWORK

Dealing with Scale: Multiscale Features

CNNs have multiple *stages* that operate at different resolutions. Attach an independent detector to the features at each level



Lin et al, "Feature Pyramid Networks for Object Detection", ICCV 2017

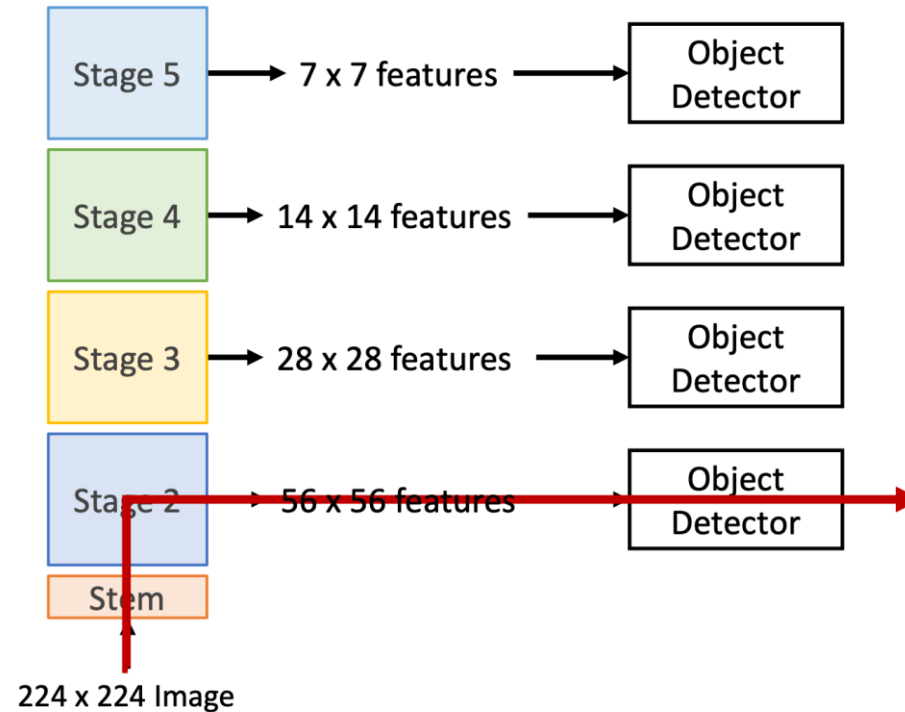


FEATURE PYRAMID NETWORK

Dealing with Scale: Multiscale Features

CNNs have multiple *stages* that operate at different resolutions. Attach an independent detector to the features at each level

Problem: detector on early features doesn't make use of the entire backbone; doesn't get access to high-level features



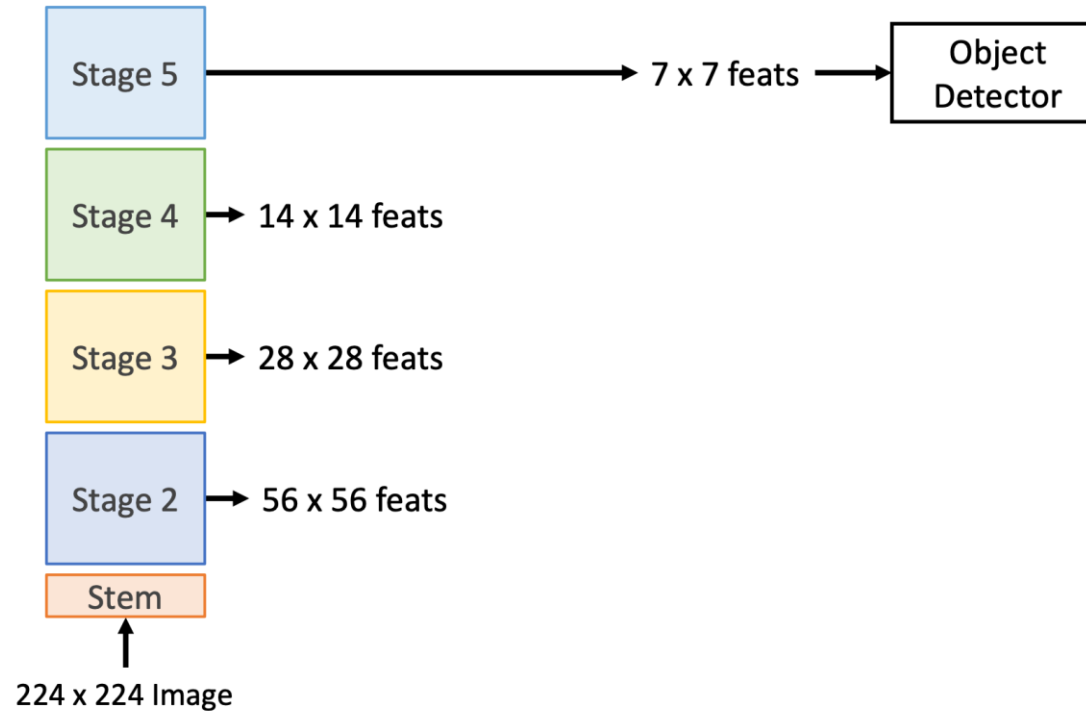
Lin et al, "Feature Pyramid Networks for Object Detection", ICCV 2017



FEATURE PYRAMID NETWORK

Dealing with Scale: Feature Pyramid Network

Add *top down* connections that feed information from high level features back down to lower level features



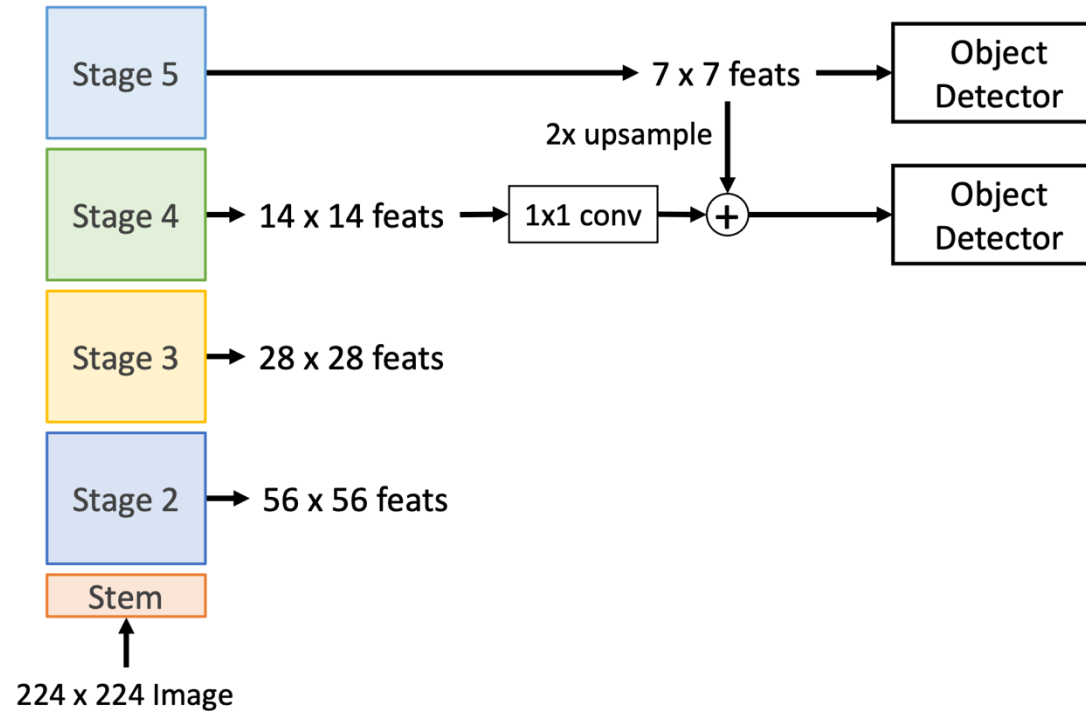
Lin et al, "Feature Pyramid Networks for Object Detection", ICCV 2017



FEATURE PYRAMID NETWORK

Dealing with Scale: Feature Pyramid Network

Add *top down* connections that feed information from high level features back down to lower level features



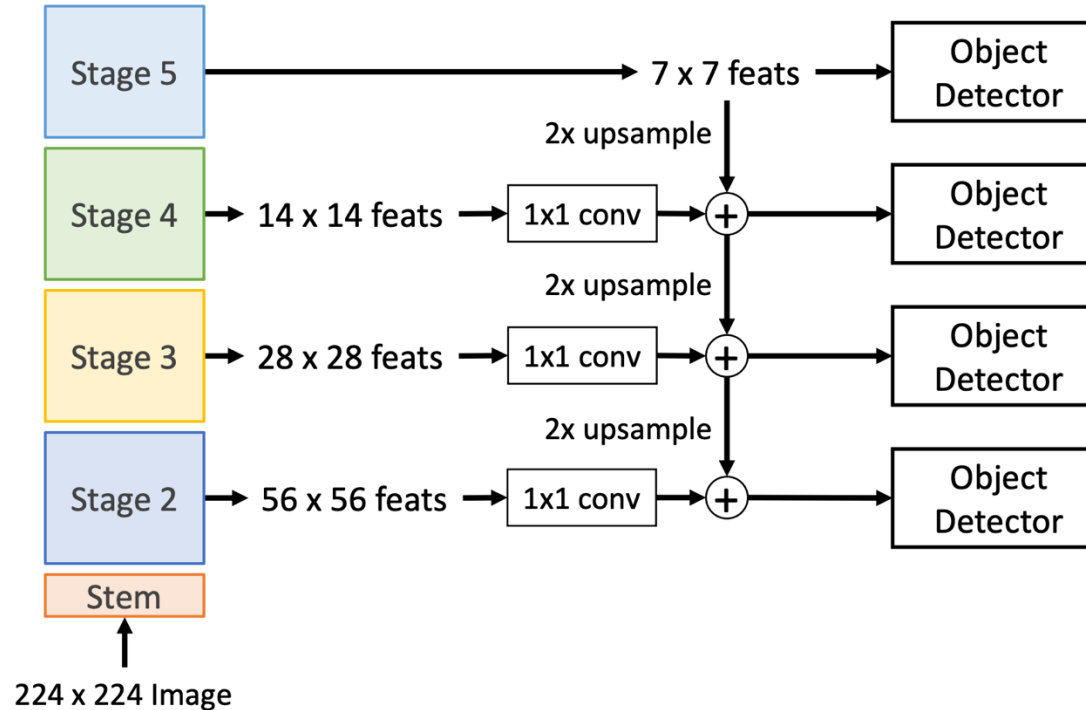
Lin et al, "Feature Pyramid Networks for Object Detection", ICCV 2017



FEATURE PYRAMID NETWORK

Dealing with Scale: Feature Pyramid Network

Add *top down* connections that feed information from high level features back down to lower level features



Lin et al, "Feature Pyramid Networks for Object Detection", ICCV 2017

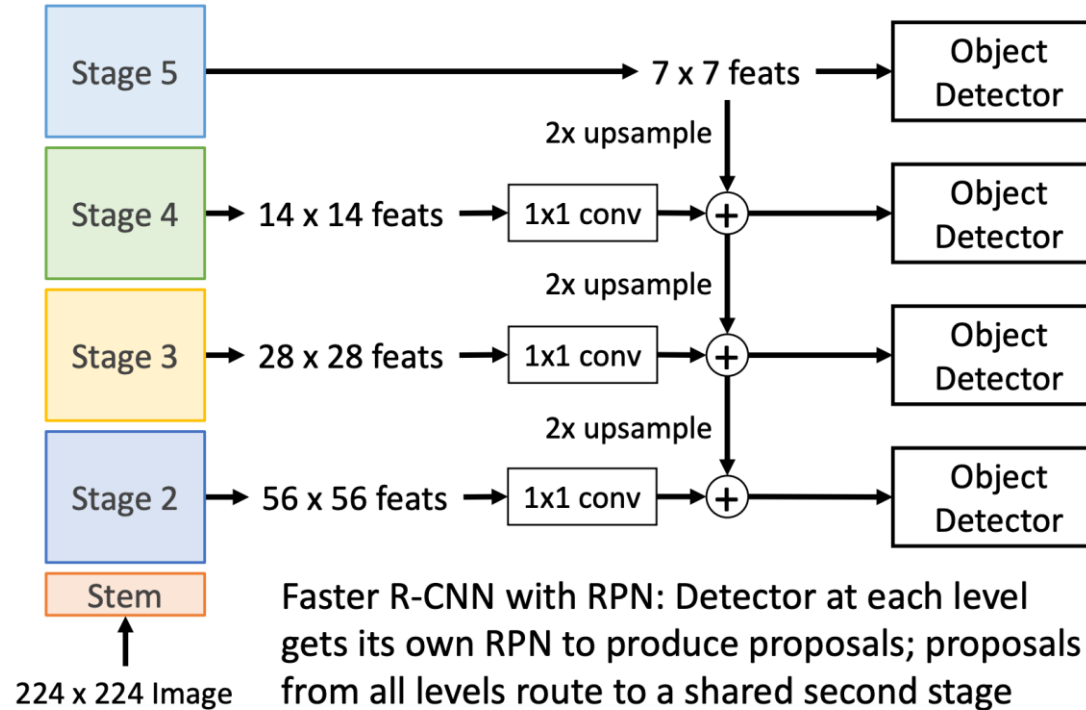


FEATURE PYRAMID NETWORK

Dealing with Scale: Feature Pyramid Network

Add *top down* connections that feed information from high level features back down to lower level features

Efficient multiscale features where all levels benefit from the whole backbone! Widely used in practice

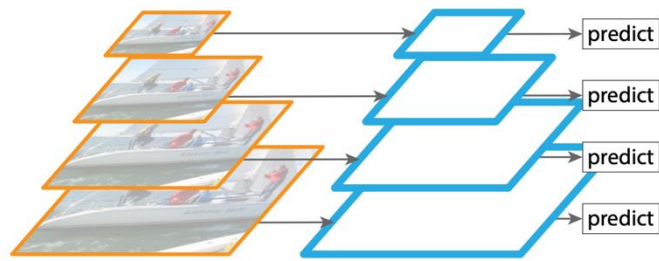


Lin et al, "Feature Pyramid Networks for Object Detection", ICCV 2017

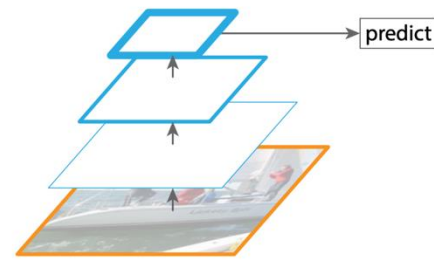


FEATURE PYRAMID NETWORK

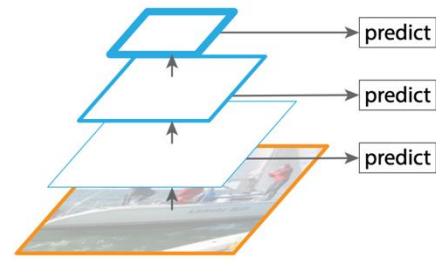
Feature pyramid



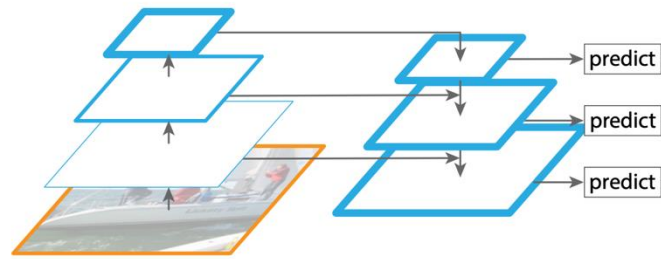
(a) Featurized image pyramid



(b) Single feature map



(c) Pyramidal feature hierarchy



(d) Feature Pyramid Network

An abstract graphic on a light gray background. Two thin, dark gray lines intersect. One line is nearly vertical, starting from the top center and extending towards the bottom right. The other line is nearly horizontal, starting from the top left and extending towards the middle right. The word "SUMMARY" is written in a bold, black, sans-serif font, positioned to the right of the intersection point.

SUMMARY

SUMMARY

1. **Faster R-CNN**
2. **Feature pyramid network**



NEXT LECTURE

Perception (5) – Single-stage object detector



Two thin, dark gray lines intersect diagonally on a light gray background. One line runs from the top-left towards the bottom-right, and the other runs from the top-right towards the bottom-left. They cross each other in the upper-left quadrant of the image.

QUESTIONS?